Score-linked practice intelligence from long-form violin training video

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High-level music training produces dense evidence: daily practice videos, repeated repertoire, changing technical constraints, score-free technique exercises, and periodic performance goals. Most of that evidence is difficult to use because it is stored as unstructured media rather than as a score-linked or pattern-linked longitudinal record. We report Curtis, a live AO Labs system that converts messy long-form violin practice videos into structured, evidence-linked practice records using active practice detection, playable audio/video evidence windows, score alignment targets, technique-pattern targets, heat-map targets, repertoire tracking, and passage-level performance observation. The checked source state at 2026-05-12 04:59 UTC used the YouTube channel `@nalalan` as the primary source, indexed 213 public videos, marked 108 as practice candidates, and stored 80 media-sample index entries. The uploaded-video ledger begins at `violin 1`, uploaded on 20 December 2025, and currently counts 44 violin-numbered or date-titled practice logs totaling 201 h 6 min of public session duration; this duration is not counted as active practice time. The daily-record state contains 37 indexed practice days, 5 audio-evidence records, 4 score-audio-only records, 1 detected-note-fragment record, 60 violin-positive sample windows, and 20 withheld non-violin windows. Total practice time is counted only from media windows where violin playing is detected; it is not tied to transcription or score matching. The current measured practice total is 33 min 6 s from 1 h 40 min of checked video, which extrapolates to an approximate 66 h 14 min total practice-time estimate across the 201 h 6 min archive until the full low-cost scan completes. All unverified room, setup, laptop, silence, or non-playing samples are withheld rather than counted as measured practice evidence. The primary interface has three top-level surfaces: a paper card, analyzed practice days, and repertoire. The analyzed-days evidence row reports total practice time first, then video checked for practice-time detection, then the full uploaded-video archive so the interface does not hide either the multi-hour corpus or the scan coverage. The analyzed-days list keeps practice days chronological and displays

exact audio-checked D4 and A4 fragments from the 2026-05-03 Scherzo-Tarantelle practice sample as detected-note evidence, not as score-location matches. Score match groups are now driven by pitch-class sequence agreement against an available score or reference sequence; the Scherzo-Tarantelle target includes a 179-note local reference-audio pitch trace so detected played notes can produce score-reference, transcription, audio, and video groups while exact measure and rhythm alignment remain pending. Each daily record withholds embedded playback, paired audio, and practice-time credit until violin playing is detected in the source window, withholds broader sheet-music notation until reliable note extraction supports either score alignment or a score-free technique-pattern record, and withholds exact score-location snippets until a played window has explicit measure agreement rather than only a source-confirmed piece title or broad pitch-sequence match. Draft notation does not render as transcription; repeated single-pitch traces such as a spurious `D D D D` stream remain out of final score-verified notation. The transcription-run PDF now lists all stored detected note series for the day, while score matching uses pitch-class sequence agreement with rhythm and octave ignored. Curtis keeps source-confirmed pieces, explicit calibration titles such as scales and arpeggios, official YouTube Data API public-reference metadata, nearest reference targets, and score context available without turning untrusted note events into accepted score evidence. Manual day-level source labels can be entered from the analyzed-day record, but these labels only name the practice source; they do not unlock score snippets, heat maps, readiness scores, or repertoire progress without matching audio and score evidence. Public labeled YouTube references seed likely reference targets but do not become repertoire, audio fingerprints, or judged performance evidence until a permitted media path and score or pattern alignment exists. Active material with no confirmed score is labeled as piece-or-exercise pending rather than forced into a wrong repertoire guess. Repertoire entries are promoted only from confirmed daily-record evidence and keep possible labels, user-rejected false labels, weak evidence, and readiness scores inferred only from piece identification separate from confirmed repertoire and judged playing quality. The system uses GPT-5.5 for vision and text review and separate audio models for identification and verification, while preserving explicit limits: Curtis admission cannot be predicted from current samples, weak audio/video evidence is marked unjudged, whole-session transcription is not yet complete, and the system reports practice signals rather than admissions probability. The contribution is an experimental method for making practice visible, searchable, measurable, and tied to actual musical evidence.

1 Introduction

Elite music preparation depends on repeated, deliberate practice and feedback over long time horizons (*1*, *2*). The useful data are not only polished performances. They include ordinary practice sessions, failed takes, camera angle limitations, recurring technical problems, and changes in focus across days. A violinist recording daily practice can therefore produce a large longitudinal dataset, but the raw dataset does not automatically become an operating record.

Curtis addresses this problem as a live software system rather than as a portfolio page (*3*). The system indexes public practice media, separates uploaded video duration from detected violin-playing practice time, records technical observations, and keeps evidence tied to clips and generated notation. It is designed for the Curtis Institute goal, but it does not claim to estimate an acceptance probability. The current role of the system is narrower: observe practice evidence, retain history, separate judged from unjudged evidence, and identify the specific passage-level blockers that can be tested in subsequent recordings.

The core pipeline is video → active practice detection → clips → music transcription → repeat grouping → score matching → heat map → repertoire update → passage-level observations. The output is intentionally small: analyzed practice days and an automatically updated repertoire list, both grounded in clips, notation, active playing time, and confidence limits.

The system is also part of AO Labs' broader progress architecture. In May 2026, AO Progress was introduced as a shared ledger for cross-application state and long-term changes across public apps, papers, CV artifacts, Curtis practice state, Imagineer state, and Relay state (*4*). Curtis supplies one stream in that ledger: timestamped media and review evidence for musical development.

2 System state

The production state uses the live backend at `https://curtis.aolabs.io/api/curtis/media-status`. The checked source state at 2026-05-09 07:16 UTC reported the values in Table 1. These are operating facts, not outcome claims.

Table 1. Curtis source state checked at 2026-05-09 07:16 UTC.

Signal	Current value
Primary source	YouTube channel `@nalalan`
Inventory	213 indexed videos
Practice candidates	108 videos
Long-form candidates	88 videos
Uploaded-video marker	`violin 1`, uploaded 20 December 2025
Title-confirmed uploaded-video ledger	44 videos, 201 h 6 min uploaded session duration
Estimated total practice time	66 h 14 min extrapolated from the checked-window active-playing ratio; approximate until the full archive scan completes
Measured practice time	33 min 6 s of detected violin-playing footage; this is independent from transcription
Video checked for practice time	1 h 40 min of the 201 h 6 min uploaded-video ledger
Unchecked archive duration	199 h 25 min still awaiting violin-playing detection
Media samples	80 indexed windows; 60 violin-positive; 20 withheld
Reviewed sections	80 sections
Curtis-focused findings	80 findings
Latest dated practice log	5-6-26
Review model	GPT-5.5 for vision/text, separate audio models for audio evidence and piece verification
Confirmed labels	5/1 Haydn Symphony No. 94 IV; 5/2 and 5/3 Wieniawski Scherzo-Tarantelle
Training anchors	3 source-confirmed practice sources
Score-reference targets	3 configured
Reference training	3 source-confirmed score targets, 22 pitch/rhythm windows, title-labeled calibration anchors for future scale/arpeggio uploads, and 4 official YouTube Data API public-reference queries kept separate from repertoire claims and audio evidence
Analyzed practice days	37 indexed days; 5 practice-time measured daily records; 5 audio-evidence records; 4 score-audio-only records; 1 detected-note-fragment record; 3 confirmed source labels with score targets, 16 pitch-sequence score groups, and 0 exact score-matched snippets
Machine pitch events	Detected-active-section onset-aware pYIN plus onset-bounded spectral pitch rescue for fast passages, audio-agreement counts, pitch-sanity filtering, repeated-pitch-collapse guarding, practice-coverage fields, title-labeled calibration handling, and fingerprints stored only when quality gates pass; broad candidate micro-transcription is withheld, while detected note series are preserved in the transcription PDF and promoted to score matches only after pitch-class sequence agreement with an available score/reference sequence
Specific observations	Rendered only when grounded in reliable score-linked evidence or repeated score-free technique-pattern evidence
Score-linked windows	16 pitch-sequence score-reference groups; 0 exact score-aligned windows
Daily evidence surface	Local sample playback only for violin-positive windows, notation withheld until reliable note/rhythm verification, score snippets hidden until played-to-score agreement, and explicit whole-session transcription limit
Curtis-level completion	Not scored when the available evidence only identifies repertoire
Current focus	Capture a playable audition-view take
Current constraint	Record 30 seconds of sustained tone
Boundary	Curtis admission cannot be predicted from current samples

2.1 Media indexing

The source inventory stores platform, title, publication time, duration, view count, candidate reasons, media kind, and blocker state. Dated long-form practice logs are treated as high-value longitudinal records because the title contains a practice date and the duration preserves session-level evidence. The uploaded-video ledger starts at the `violin 1` marker and counts only violin-numbered or date-titled practice logs, so unrelated long public videos are retained in inventory without being silently counted as violin practice. This ledger is not the active-practice total. Active practice is measured only after media processing identifies windows where violin playing is actually occurring.

2.2 Review evidence

The review layer stores section-level findings with a dimension, evidence source, judgment, and practice constraint. The current dimensions include tone, intonation, shifts, time, articulation, and audition delivery. Findings can be strong, needing work, or unjudged. The unjudged state is part of the system design: if the sound is not available, the hand is obscured, the camera angle hides a transition, or still frames do not support a timing claim, the system should not convert weak evidence into a confident critique.

3 First readout

The first live readout is experimental and intentionally conservative. The strongest immediate value is that the system has already converted a large, timestamped practice archive into a queryable corpus with captured media samples, stored section reviews, and title-confirmed practice records. The home screen has three top-level surfaces: a paper card, analyzed practice days, and repertoire. The first evidence row now reports total practice time first, then the video already checked for violin-playing, then the 201 h 6 min YouTube practice archive; this prevents the public practice ledger from being confused with either transcription coverage or uploaded-video duration. Each analyzed day groups same-day videos, preserves uploaded duration, and names confirmed or uncertain piece evidence without promoting guesses. Playable local media samples, paired audio/video evidence windows, and practice-time credit are now withheld unless violin playing is detected in the source window. The owner-sync sampler now reuses locally cached captured media, uploads cached violin-positive windows before blind new captures, expands around already-found violin-positive windows, and can switch into full-check mode to walk every 90-second window in order instead of staying limited to anchor samples; the live analyzed-days surface currently exposes violin-playing practice-time evidence for 5/1, 5/2, 5/3, 5/5, and 5/6. Broad sheet-music notation is also withheld until reliable note/rhythm verification, so audio evidence cannot appear as a failed transcription. The current visible notation is limited to exact audio-checked D4 and A4 fragments from 5/3, paired with generated local clips for the same source sample windows. Actual score images are shown only inside matched evidence groups: a source-confirmed piece title creates a score target, and a detected pitch-class sequence match can show the score page beside the played clip and transcription while exact score-location agreement remains pending. The daily record includes a compact source-label field so a missing or wrong day title can be corrected without letting that correction masquerade as score alignment. The analyzed-days surface no longer promotes broad candidate transcription while candidate micro-transcription fails audible match review; those windows remain hidden support data or audio evidence only. Active scans with no confirmed piece are treated as possible repertoire or score-free technique exercise evidence rather than as failed score matching. The repertoire surface is evidence-backed: an entry appears only when a daily record carries confirmed source evidence, and each entry keeps practice dates, clips, score targets, practice time when measured, and current blocker limits. Generic etude/caprice evidence remains outside the piece-title field, and possible or uncorroborated repertoire labels remain checkable against the source window. These statements are not generic pedagogy; they are extracted from stored review findings, sample indexes, source corrections, public-domain score targets, pitch-quality checks, and the uploaded-video ledger.

Table 2. Current evidence classes and limits.

Class	Stored signal	Limit
Audio	Tone, intonation, shifts, time, articulation	Some audio reviews fail or cannot inspect the excerpt cleanly
Video	Setup, posture, bow lane, visible hand/bow evidence	Still frames can hide motion, sound, and transition accuracy
Metadata	Dates, duration, channel, view count, practice-candidate labels	Metadata cannot judge playing quality by itself
Progress plan	One focus, practice constraint, short session plan	The plan is a next experiment, not an outcome prediction

Current transcription limit. Machine note output now has three display gates. First, the media window must be classified as violin-positive before it can appear as playable evidence or count toward practice time. Second, the note stream must audibly match the paired audio before any notation can render. Third, repertoire passages need score alignment, while score-free exercises need repeated-pattern grouping before they can become full practice transcription. Source-confirmed score targets are also display-gated: an actual score snippet is hidden unless the corresponding played window has explicit score-location agreement. Unverified room, setup, laptop, silence, or non-playing samples remain withheld, and unverified pitch traces remain hidden support data rather than visible notation. A spectral-onset rescue path estimates pitch inside short onset-bounded violin segments when the pYIN path collapses, while the repeated-note trace guard prevents fast or complex playing from becoming a dominant repeated pitch, records the quality mode, and keeps unusable traces out of accepted score evidence. The current micro-transcription path stores candidate short contiguous runs, but it does not render them as sheet music or count them as solved transcription until the displayed notes pass paired-audio listening review. The transcription-run PDF preserves all stored detected note series for the day. Score matching uses pitch-class sequence agreement, ignores rhythm and octave for the initial grouping step, and can promote detected Scherzo-Tarantelle fragments only when the played pitch-class sequence appears in the local reference-audio pitch trace or another available score/reference sequence. A separate audio-checked fragment gate can render a note when pYIN and YIN agree on the same semitone over a short stable window; the present live examples are D4 and A4, not a full passage and not a score-location match. Full-session transcription remains incomplete until every playing section in the long-form practice videos is extracted, verified as violin-positive, verified against the audio, and aligned either to a score section or to a repeated technique pattern.

4 Discussion

Curtis Media Review is intentionally conservative. It does not rank the player against applicants, infer admissions probability, or convert every video into a claim. Instead, it keeps a running technical record. The sampler now treats violin presence as a hard evidence boundary: playable clips and practice-time credit appear only after violin playing is detected in a media window, while notation waits for reliable note/rhythm verification and either score alignment or score-free pattern grouping. This makes the system useful even before it becomes musically sophisticated: it can preserve practice volume, surface recurring constraints, identify missing evidence, and make subsequent recordings easier to compare against earlier ones.

The main open problem is repertoire identification, score-free exercise representation, and evidence quality. YouTube metadata can identify candidate videos and public labeled reference sources, and the current owner-sync path can provide direct audio/video samples, but unclear camera framing, missing score/title context, improvised technical patterns, and short excerpts still prevent reliable piece naming. The backend therefore keeps generic labels out of the piece-title field, stores them as candidate evidence, requires source-window evidence for confirmed titles, and marks weak evidence explicitly. The current training ledger separates source-confirmed labels from title-labeled calibration uploads, public YouTube reference metadata, independent audio-title matches, score-free or piece-pending active windows, failed unverified machine pitch/rhythm windows, failed pitch-collapse windows, withheld candidate micro-transcription windows, audio-checked single-note fragments, detected note series, and score-aligned windows. It also stores reference targets for the three confirmed source labels across two works so later passes can ask for section-level matching instead of repeating broad title guesses. The sampler classifies candidate windows using active audio, violin-range voiced pitch, onset density, pitch-class diversity, spectral features, and minimum practice-duration gates before a sample can become visible evidence. The transcription path converts sampled media to mono audio, finds audio-active playing sections, runs pitch tracking only on those sections, normalizes and harmonic-isolates each section for pitch tracking, estimates monophonic violin pitch with an onset-aware pYIN pitch track bounded to the violin range, detects onset boundaries so repeated attacks on the same pitch are not silently merged into one sustained note, runs a second spectral pitch estimate inside short onset-bounded segments when fast playing defeats pYIN, compares pitch diversity across candidate note streams, checks whether selected note events agree with an independent spectral or pYIN/onset detector, rejects out-of-range pitch artifacts, requires sustained pitch changes for pitch-track segmentation, removes very short low-confidence pitch blips, marks isolated octave-flip corrections as uncertain, rejects repeated-pitch collapse, estimates tempo, stores detected note series for audit/PDF review, and builds pitch-class, interval, rhythm, and contour fingerprints only when quality gates pass. Daily records now carry transcription quality, material status, and practice-coverage states, and the primary interface withholds media playback unless the source window is violin-positive, withholds broad machine notation until note/rhythm verification, displays only audio-checked fragments when the stricter gate passes, and withholds score match groups unless the detected pitch-class sequence appears in an available score/reference sequence. A stronger system should align repertoire events against a symbolic or optical score representation, group score-free technical patterns by repeated note/rhythm fingerprints, identify movement sections or measure ranges when a score exists, compare against reference recordings available through permitted media, and then judge articulation, timing, tone, and tempo against the aligned passage or repeated pattern.

The broader experiment is whether a practice system can make musical development legible over time. A useful Curtis system should show whether tone, intonation, rhythm, articulation, shifts, memory, endurance, and audition delivery are improving across dated recordings. It should also preserve the source evidence behind each claim so that the dashboard does not become motivational text detached from the actual playing.

5 Materials and methods

Backend

The backend is a FastAPI service deployed on Railway. It exposes `https://curtis.aolabs.io/api/curtis/media-status` for compact state, `https://curtis.aolabs.io/api/curtis/ops-check` for fuller operational state, local stored sample media for embedded playback after the media is violin-positive, sample indexes for duplicate-window avoidance, and run endpoints for scans, media probing, analysis, and coaching. Persistent runtime state stores sources, inventory, review findings, media samples, analysis runs, and scan history.

Source scan

The default public source is `https://www.youtube.com/@nalalan`. The scanner queries public video metadata, classifies videos by title, duration, and candidate reasons, and stores inventory records. Dated practice logs and long-form videos are treated as practice candidates. The media probe now keeps one early baseline window but then probes deeper quarter, midpoint, five-eighths, three-quarter, and tail windows before spending additional passes on early setup probes, and each stored sample is keyed by video identifier plus window start so later passes can expand coverage without overwriting earlier evidence. When configured above the anchor-sampling threshold, the probe and owner-media helper generate dense 90-second windows from the start of each video through the latest valid window so the full practice archive can be checked without the earlier retention cap. Each sample is then classified by the violin-presence sampler before it can count as active practice evidence. The owner-media helper also checks its local cache of already captured browser samples, reuses classifier results when the file signature and classifier version match, prioritizes unsynced violin-positive cached windows before capturing more blind windows, and expands around already-found violin-positive windows before returning to blind sampling. The practice-hours total uses a stricter title-confirmed ledger from `violin 1` onward: violin-numbered videos and date-titled logs count toward the top-line session duration, while unrelated long videos remain indexed but outside the practice-hours total. The scan does not itself inspect performance quality.

Model review

The model layer reviews sampled media sections where usable excerpts exist. GPT-5.5 is used for text and visual review, and separate audio models are configured for audio review and piece-title verification. Findings are sanitized so weak evidence terms such as obscured, not audible, or no clear evidence produce an unjudged state rather than an unsupported critique. Piece labels are also sanitized: coach reviews can store candidate evidence but cannot create confirmed repertoire titles, and the piece-identification layer must provide source-window evidence plus independent verification before it updates the current piece list. Same-video consensus can compare multiple sampled windows before accepting a title; for long date-stamped practice captures with later samples, setup or non-playing windows are excluded from confirmation, and confirmed active titles are versioned so older labels are rechecked when source rules change. User-rejected false labels are excluded from later passes only for the source where the correction applies, so later videos can still identify that repertoire when present. A source correction also invalidates older confirmed labels for that same source instead of promoting the next stale candidate, and repeated corrected false labels keep that source out of the confirmed repertoire list until a stronger acceptance path exists. Rejected proposed titles, top candidates, and verification titles are scrubbed from current API output, not only from confirmed piece records. The 5/1 forensic pass rejected prior Paganini, Bruch, Mendelssohn, Brahms, Sibelius, Tchaikovsky, Wieniawski, Saint-Saens, Ravel, Bazzini, Ernst, Carmen, Zigeunerweisen, Bach Partitas, Kreisler Praeludium and Allegro, Sarasate family labels, Ysaye Sonata No. 3 Ballade, Mozart K. 216, Glinka Ruslan and Lyudmila, Strauss Till Eulenspiegel, Strauss Don Juan, Smetana Bartered Bride, Prokofiev Classical Symphony, Dvorak Carnival Overture, Shostakovich Symphony No. 5, Ravel Bolero, the first broad orchestral candidate batch, and the later overture, dance, and pops-orchestral candidate batch for that source. Alan then supplied the ground-truth source label: Haydn Symphony No. 94, IV. Finale, Violin I part. The system now treats that label as human-confirmed for 5/1, stops injecting new seeded guesses for the same source, and keeps Curtis-level completion unscored until judged playing evidence supports a score. Alan also supplied the 5/2 source label, Wieniawski

Scherzo-Tarantelle, Op. 16, and then confirmed that all violin footage on 5/3 was the same piece. The training state now records all three source labels as anchors and reports zero score-aligned windows until pitch/rhythm extraction has matched a source window to a score section. Transcribed samples can now produce learned pitch/rhythm fingerprints only after their media window is violin-positive, and visible notation requires note/rhythm verification. Future samples can receive a nearest-source hint when their extracted notes and rhythms resemble a prior confirmed source, an explicit title-labeled calibration upload, a symbolic reference, or a public labeled YouTube reference indexed through the official Data API; if the extracted pattern does not belong to a known score, it remains piece-or-exercise pending rather than being forced into repertoire. Public reference metadata can improve candidate vocabulary and target selection, but it does not become an audio fingerprint until a permitted media source exists. Possible labels and model-only verified titles remain uncertain daily evidence unless a source label is confirmed, so future misses appear as missing evidence rather than as wrong repertoire. Piece-identification output does not create a Curtis-level readiness percentage by itself; readiness scoring requires judged playing evidence. Dated practice entries are keyed by the source-video date rather than the later processing time, so historical uploads cannot masquerade as today's practice state. The owner-browser sync uses the sample index to prioritize uncaptured public YouTube windows rather than re-uploading already reviewed excerpts, and the piece-identification pass can test multiple active moments inside each captured sample before withholding the title. The daily-record builder joins the strict `violin 1` ledger to violin-positive media samples, hidden machine note/rhythm events, practice-time evidence, clips, score packet support, repeated-fragment or repeated-pattern heat maps, score-aware read fields, and observation records. Repertoire entries are built only from confirmed daily-record evidence and keep progress percentages unassigned until clips, notation, and score or pattern alignment support them.

Progress ledger

AO Progress tracks Curtis as part of a larger multi-application state record. The first progress snapshot included Curtis home, Curtis media state, other AO Labs apps, Imagineer state, Imagineer paper, Relay state, and public project surfaces. Curtis contributes dated practice inventory, current focus, review status, and future trend signals.

6 Limitations

The current system cannot predict acceptance at Curtis. It also cannot replace a musician, teacher, jury, studio lesson, or audition process. It can be wrong when the source media are incomplete, when audio is unavailable, when video frames hide the physical action, or when model review fails. The broad micro-transcription gate is currently held closed: candidate note runs can be stored, but no displayed passage transcription should be treated as solved until its notes audibly match the paired clip. The only current displayed notation is two strict audio-checked note fragments, so the useful claim is limited: Curtis Media Review turns practice media into a persistent, source-bounded record that can support iteration over time.

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